# Virtualization Technology

Virtualization is a foundational technology underlying modern cloud computing, allowing a single physical computing resource to be divided into multiple virtual resources, each functioning as an independent system with its own operating system and applications. At the core of virtualization is the hypervisor, a software layer that manages and allocates physical hardware resources, such as processing power, memory, and storage, among multiple virtual machines running on the same physical server. Type 1 hypervisors, also known as bare-metal hypervisors, run directly on physical hardware without requiring an underlying host operating system, generally offering superior performance and are commonly used in enterprise and cloud computing environments. Type 2 hypervisors run as an application on top of an existing host operating system, offering greater flexibility and ease of use, though typically with somewhat reduced performance compared to bare-metal alternatives, making them more common in development and testing environments. Virtual machines created through virtualization behave as complete, independent computing systems, each with its own virtual processor, memory, storage, and network interfaces, allowing multiple isolated environments to run simultaneously on shared physical hardware without interfering with one another. This technology provides significant benefits, including improved hardware utilization, since multiple virtual machines can share the resources of a single physical server that might otherwise be underutilized running a single application. Virtualization also enables important capabilities such as live migration, allowing virtual machines to be moved between physical servers with minimal or no downtime, supporting maintenance activities and load balancing without disrupting running applications. Beyond traditional virtual machines, containerization technology, exemplified by platforms such as Docker, has emerged as a lighter-weight alternative, packaging applications along with their dependencies into portable containers that share the underlying operating system kernel, offering faster startup times and more efficient resource utilization compared to traditional virtual machines.